

Business Problem Statement

A retail company collects large amounts of customer shopping data from daily transactions but lacks a clear understanding of the factors influencing customer purchasing behavior. The management team wants to gain deeper insights into customer preferences, spending habits, product performance, and seasonal trends in order to make better business decisions.

The company is interested in identifying patterns related to customer demographics, purchase frequency, payment methods, product categories, and customer engagement. Understanding these trends can help improve marketing campaigns, optimize inventory planning, enhance customer experience, and increase overall revenue.

As a Data Analyst, the objective of this project is to analyze customer shopping behavior and transform raw transactional data into meaningful business insights.

Business Question

How can customer shopping data be used to identify purchasing trends, understand customer behavior, and support data-driven business decisions that improve sales and customer satisfaction?

Project Deliverables

1. Data Preparation & Cleaning (Python)

- Load and inspect the dataset.
- Handle missing values and data quality issues.
- Perform data cleaning and transformation.
- Prepare the dataset for analysis and visualization.

2. Business Analysis (SQL)

- Analyze customer purchasing patterns.
- Evaluate sales performance across product categories.
- Identify customer segments and spending behaviors.
- Generate insights using SQL queries.

3. Dashboard Development (Power BI)

Build an interactive dashboard to visualize:

- Sales by Category
- Customer Demographics
- Seasonal Trends
- Payment Method Preferences

- Revenue Insights
- Customer Purchase Behavior

4. Business Insights & Recommendations

Summarize key findings and provide recommendations that can help the company:

- Improve customer engagement
- Increase sales performance
- Understand customer preferences
- Support strategic business decisions

5. Project Documentation & GitHub Repository

Maintain a well-structured repository containing:

- Python Scripts
- SQL Queries
- Power BI Dashboard
- Dataset
- Project Documentation